

FORMULABENCH

FormulaBench

Auditable Excel formula generation

Research: Excel Formula Generation

Encode x Ylookup Rebuild Private Markets Hackathon

```
=IFERROR(  
  INDEX>Returns!$H:$H,  
  MATCH($B9>Returns!$B:$B,0)),  
  0  
)
```

ILLUSTRATIVE FORMULA

400-task public benchmark plus a controlled 15-task Tinker checkpoint comparison

Public self-evaluation: 133 of 400 tasks passed

The run recalculated every workbook in LibreOffice before using the organiser-supplied evaluator.

33.25%

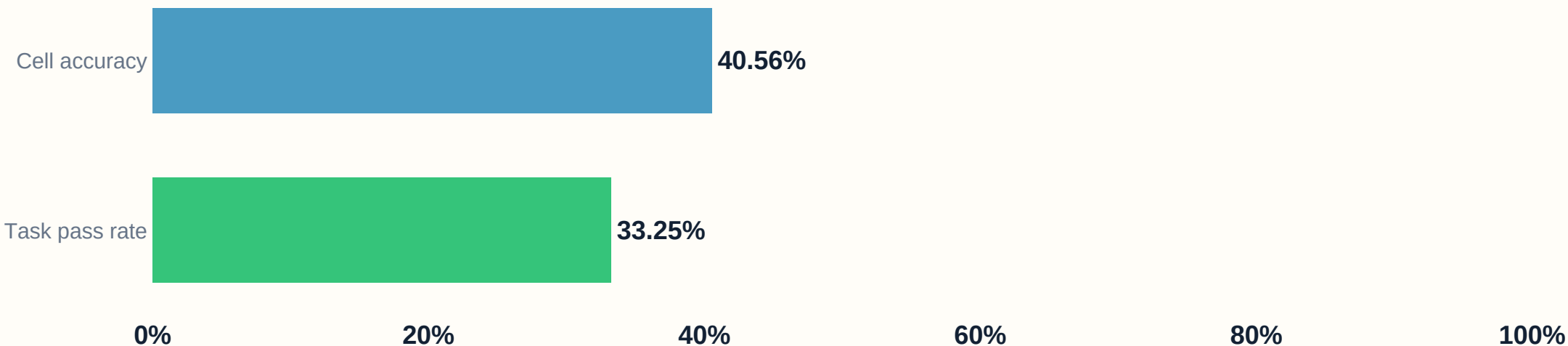
task pass rate

40.56%

cell accuracy

400/400

All 400 tasks evaluated, with no missing tasks or evaluator errors.



120,824 of 297,882 target cells matched the reference workbook.

116 of 120 correct cells still produced a failed task

Two retained benchmark tasks show the difference between cell accuracy and a complete pass.

TASK 54513

FULL PASS

1 / 1

target cell correct

\$34.99 with a 55% discount

Sheet1!F8

=C8*(1-E8)

15.7455

exact value

15.75

displayed value

TASK 13-1

FULL TASK FAILED

116 / 120

target cells correct

96.67% cell accuracy

Merged duplicate records, then sorted the output.

Four amounts differed. D32 is the final total.

Cell	Expected	Produced
LISTS!D25	1,990	2,020
LISTS!D30	534	394
LISTS!D31	401	276
LISTS!D32	6,495	6,260

The evaluator checks every target cell and grants a task pass only when the complete workbook answer is correct.

Plausible formulas fail in four different ways

A workbook can open successfully while still producing the wrong investment result.

01

Context

The relevant worksheet, source range or existing formula never reaches the model.

02

Coverage

A response misses required cells, writes to the same cell twice or changes cells outside the target.

03

Workbook

A formula introduces an unsafe reference, invalid type or broken workbook package.

04

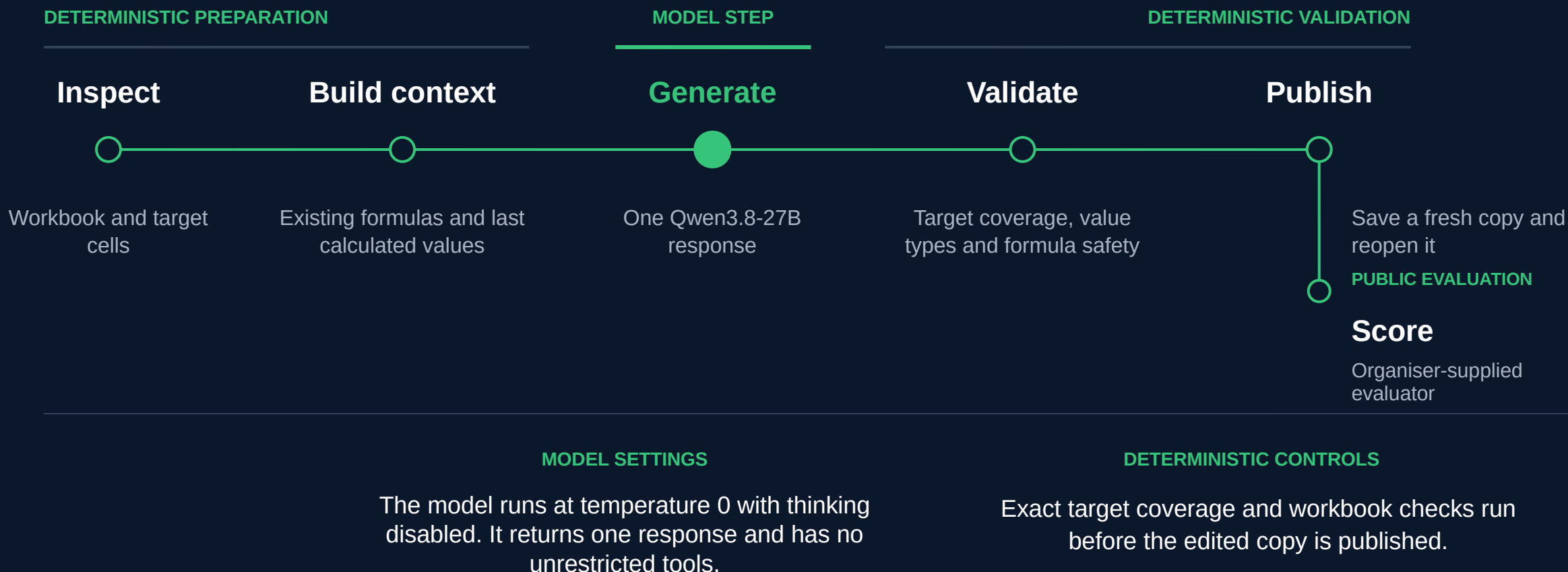
Semantics

The formula runs but returns the wrong value or applies the wrong transformation.

Execution and correctness require separate checks.

Current FormulaBench workflow

FormulaBench requests one logical model sample between deterministic preparation and validation.



Workbook checks before publication

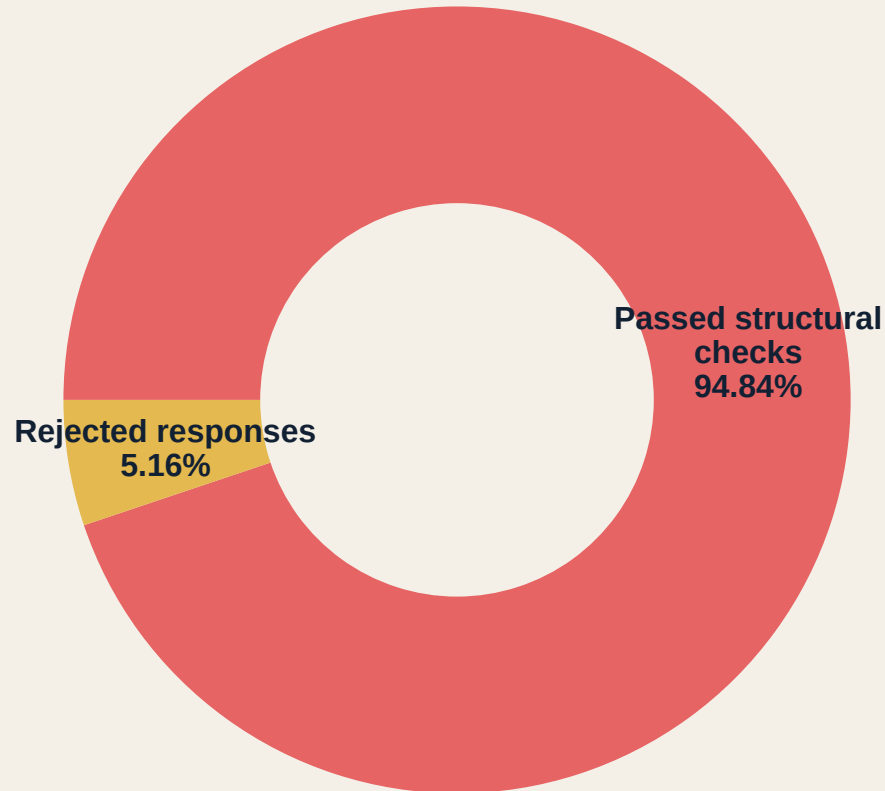
FormulaBench publishes an edited workbook only after every check passes.

Check	Required evidence	Failure behaviour
Sheet + A1	Every target carries an explicit worksheet and cell address.	Reject the full response
Exact coverage	Finite targets contain no gaps, duplicates, overlaps or extra assignments.	Reject the full response
Typed values	Dates and datetimes use explicit Excel value types rather than locale-dependent text.	Reject invalid schema
Formula gate	External paths, URIs, DDE and high-risk executable or data functions remain blocked.	Reject unsafe formula
Save and reopen	The system saves a fresh copy, reopens it and verifies the target cells.	Return the untouched workbook

Rejected responses leave the supplied workbook unchanged.

Most wrong cells were in structurally valid responses

Responses that passed structural checks contained 94.84% of wrong cells. Rejected responses contained 5.16%.



94.84%

221 structurally valid responses still contained incorrect cells.

84.76%

of all wrong cells came from four transformations with more than 10,000 target cells.

Context truncation pass rate: 24.10%

No truncation pass rate: 41.95%

This association does not establish causation.

Organiser reference and FormulaBench use different test conditions

The base model is the same. The inference setup and requested output differ.

≈59%

organiser reference

33.25%

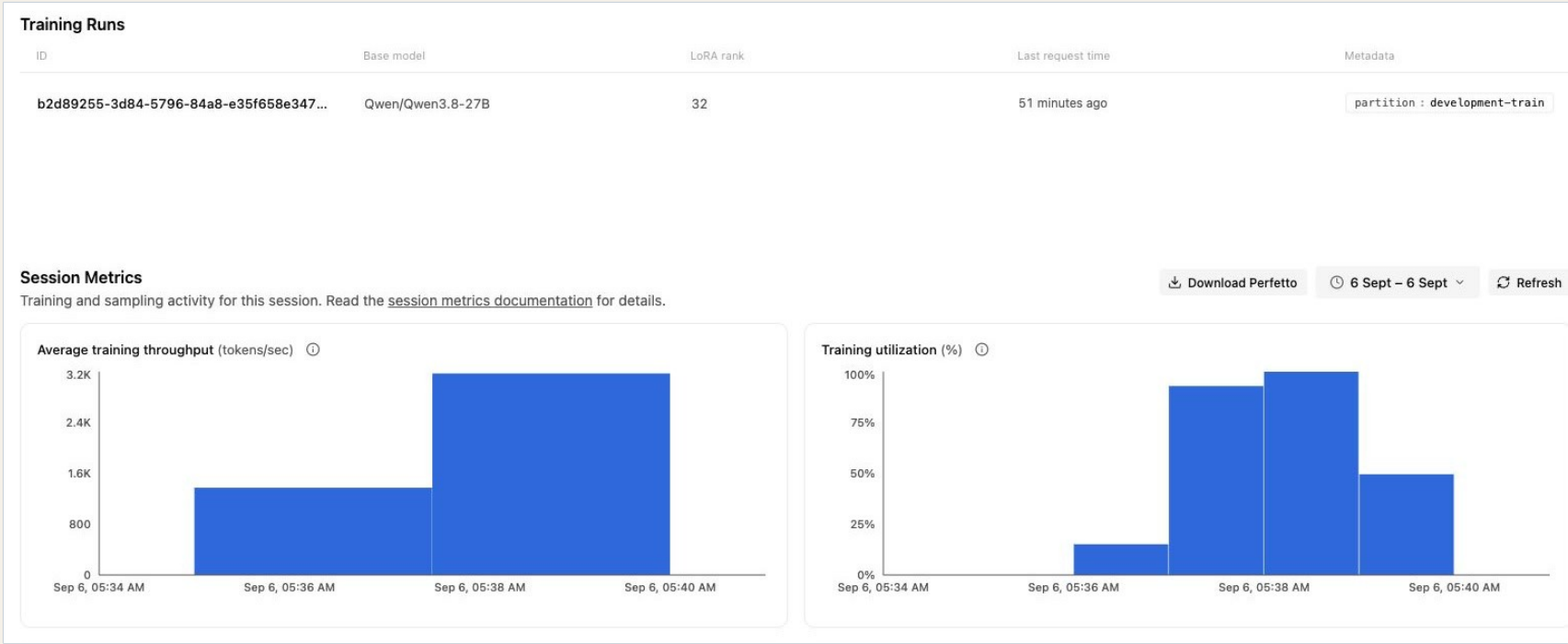
FormulaBench public self-evaluation

System	Inference mode	Requested output	Validation evidence
Organiser reference	xhigh reasoning inferred from the published runner and cookbook	Final scalar values	≈59% reported. Run traces and renderer metadata not published
FormulaBench	One native Tinker sample at temperature 0 with thinking disabled	Executable formulas and typed transformations	Organiser-supplied evaluator. Exact coverage checks and full traces are available

The published evidence does not isolate the effect of any one component.

Tinker LoRA training session

Provider evidence for the completed development run on Qwen3.8-27B.



59
training examples

15 / 15
optimiser steps completed

468,690
training tokens

The charts record training activity. Tinker did not export one exact average throughput or utilisation figure, and neither metric measures workbook correctness.

Validation NLL decreased from 0.060939 to 0.015557. This measures label prediction, not spreadsheet accuracy.

The LoRA checkpoint passed two more of the same 15 tasks

Base and checkpoint used the same prompts, settings, workbook pipeline and organiser evaluator.

7 / 15

base model

9 / 15

LoRA checkpoint

+13.33 pp

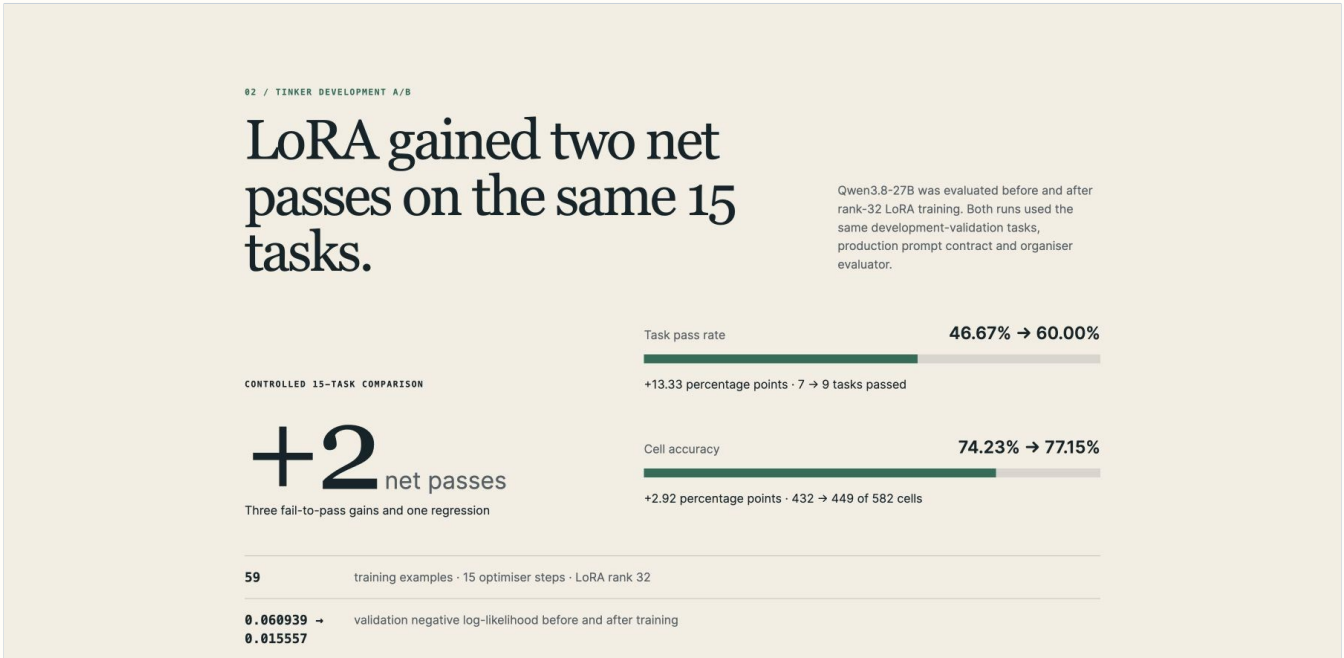
task pass rate, 46.67% and 60.00%

432 / 582

449 / 582

correct cells increased by 17

Three failures became passes. One previous pass regressed.
Accepted predictions increased from 14/15 to 15/15.



The image is the public result summary, not a separate evaluation.

Development checkpoint selection only. One sample per task in each arm. Six examples above 500 target cells were excluded. The frozen 400-task pass rate remains 33.25%.

Next architecture: separate formula generation from large transformations

Formula tasks use the model. Large transformations use fixed operations.

FORMULA TASKS

Generate a formula

Recalculate locally and run task-specific checks before publishing the workbook.

LARGE TRANSFORMATIONS

Apply a fixed operation

Use rule-based code for filtering, sorting, appending, grouping, deduplicating and clearing.

LIVE RESEARCH DEMO

masterasnackin.github.io/FormulaBench/

PUBLIC REPOSITORY

github.com/MasteraSnackin/FormulaBench

The organiser's private benchmark is the unseen evaluation.